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Enhancing solar thermal collector systems for hot water production through machine learning-driven multi-objective optimization with phase change material (PCM)

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ABSTRACT

Energy storage and supply in solar thermal collector systems rely heavily on phase change materials (PCM). It is vital to establish the appropriate values for energy discharge time (t_{PCM}) and net amount of stored energy (Q_{net}) within the PCMs in order to increase the efficiency of solar thermal collectors. For this study, we implemented a multi-objective evolutionary algorithm based on decomposition (MOEA/D) for the optimization process using the MATLAB simulator software. The simulation findings show that, under ideal circumstances where t_{PCM} is large and Q_{net} is low, these two goal functions are inversely connected to one another. According to an analysis of how input parameters affect the objective functions, the mass of PCM is minimized and the mass flow rate of the input water is increased under ideal circumstances. The relationship between the tube's inner diameter and the t_{PCM} objective function shows that an increase in the tube's diameter corresponds to an increase in the energy discharge time. The amount of stored energy in the PCM increases when the system parameters are held constant while the tube diameter is increased. Additionally, we investigated the effects of several solid PCM materials on the discharge time and the energy stored.

1. Introduction

Many dedicated researchers have conducted in-depth analyses to determine whether it is feasible to employ renewable energy sources and energy storage technologies to reduce the consumption of fossil fuels [1–4]. Solar radiation may be collected using photovoltaic cells or solar collectors, making it one of the most widely available forms of renewable energy [5]. Solar thermal collectors are effective at gathering solar energy and converting it into usable energy for a number of uses, such as heating air for HVAC systems, water for hygiene, or even storing solar heat for use as a passive heating system in buildings [6–9]. Phase change materials (PCMs) offer a very practical alternative for storing heat, especially for low-temperature applications like dwellings [10–13] and high temperature application (e.g. power generation) [14,15].

Numerous studies have examined the viability of PCMs as a fundamental form of energy storage, and all have demonstrated the enormous improvements that PCMs may bring to energy systems [16–19]. To attain the best performance in these systems, energy optimization approaches are widely utilized. These methods use a variety of objective functions that consider heterogeneous elements including energy, environment, and economy [20,21]. Numerous investigations on PCM

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Nomenclature

T_a	Ambient temperature [°C]
A	Area [m^2]
C_P	Specific heat of air and water at constant pressure [kJ/kg.K]
$\dot{m}$	Mass flow rate [kg/s]
t	Time [h]
T	Temperature [°C]
h	Height
D	Diameter
W	width
U	Overall heat transfer coefficient [kW/m^2K]
t_{PCM}	discharge time
Q_{net}	quantity of stored energy
F	*Feed water mass flow rate [kg/s]*
C_P	*Specific heat of air and water at constant pressure [kJ/kg.K]*
$\dot{Q}_u$	*collector heat gained, (kW)*
$\dot{m}$	*Mass flow rate [kg/s]*
N	number of covering glasses
ε_p	plane emission intensity
ε_g	glass emission intensity (0/88)
β	collector gradient
h_w	coefficient of wind heat transfer
$\dot{Q}$	Heat transfer rate [kW]

Subscripts

N	Number
i	in
e	out

Abbreviations

PCM	Phase Change Materials
MOEA/D	multi-objective evolutionary algorithm based on decomposition

Greek symbol

φ	plane angle

energy storage system optimization methods have been carried out. For instance, an exergoeconomic optimization of a solar thermal energy system was performed by Aghbalou et al. [22]. The system was made up of a solar collector and a rectangular water tank with a PCM casing that served as a thermal energy storage system. Their investigation into how convection and conductivity impact heat transfer offered a mathematical justification for the PCM's melting. Guo et al. [11] focused on getting the solar collector area and storage tank of a system with short-term heat storage for space heating applications to fit together as well as feasible. They were able to establish an ideal ratio between the collection area and storage tank volume.

An integrated optimization method for using PCMs in thermal storage devices was presented by Lin et al. [23]. The two performance measures that made up the study's aim function were the average heat transfer efficacy and effective PCM charging time. The use of a genetic algorithm to determine the optimum Pareto front and a method for making decisions with numerous criteria allowed the system's efficiency of heat transmission to rise from 44.25 % to 59.29 %. Furthermore, the real PCM charging time increased from 4.53 to 6.11 h. Heat storage units are used in solar energy storage systems to combat the variability and unpredictability of solar energy. These systems use PCMs as their storage medium. Nekoonam and Roshandel [24] have quantitatively investigated a system consisting of a PCM-based cascaded energy storage unit and an energy collector. They demonstrated that the various-PCM energy storage unit's charging effectiveness is increased by a minimum of 7 % in its ideal configuration. Lin et al. [25] carried out a study aimed at optimizing an experimental research and two-level model-based approach were used to integrate a solar photovoltaic thermally collector with thermal energy storage in phase change materials. According to the experimental results, regulating the air flow rate at a particular solar radiation level resulted in the solar-powered thermal collector combined with PCM thermal power storage operating at its maximum efficiency.

Mahfuz et al. [26] analysed the performance of a shell and tube thermal energy storage system using paraffin wax as the phase change material, for a solar water heating application, due to its significant capacity for storing energy and isothermal discharge and charging performance. Huang et al. [27] developed a novel solar energy storage structure and utilized a combined approach of numerical and experimental analyses to evaluate the performance of phase change materials. The findings revealed that the PCM in the proposed energy storage structure exhibited excellent performance, which was consistent with the numerical model. Asgharian and Baniasadi [28] conducted a comprehensive literature review to compare and evaluate various mathematical models, numerical methods, and thermodynamic analyses for phase change material-based systems in various solar energy storage applications. The study primarily focuses on modeling and simulation of such systems to analyse their accuracy and efficiency. Shamsi et al. [29] established a model for Cascade Latent Heat Storage that blends a genetic algorithm technique with a characteristics-based strategy to fully utilize a sequential multi-layer Phase Change Material thermal energy system for storage. The study aims to enhance the efficiency and capacity factor of the thermal heat storage system, which is a crucial component in solar power plants for reducing power generation costs. The study found that applying the genetic algorithm and the characteristic-based strategy to optimize the system led to a considerable improvement in system performance. Specifically, the optimization yielded a 5 % increase in energy discharged over an 8-hour charge-discharge cycle, and a 5.12 % increase in energy stored during a 4-hour charge period, surpassing the performance a storage device with a single PCM.

Fossil fuels are limited and will eventually run out, or there will come a time when it will be too expensive to extract what is left. Also, fossil fuels cause air, water, and soil pollution, producing greenhouse gases that cause global warming. In this situation, renewable energy sources are clean alternatives to fossil fuels. Nowadays, researchers pay attention to renewable systems for producing different energies, including electricity, cooling, heating, hydrogen, and fresh water, and to overcome the crisis of energy shortage, they pay attention to renewable systems, including solar, wind, and earth systems. Heat is a necessary thing that currently many researches and studies have been done on solar systems, but more complete research is needed to complete renewable studies [30–43].

The use of hot water thermal storage systems is very widespread; Because fossil resources are limited and energies must be compatible with the environment in such conditions. This method is an efficient solution for energy storage that has a high capacity to store energy and supply it at a constant temperature (isotherm mode). To store energy, so-called phase change materials (PCM) are used, which have the task of storing energy in the form of latent heat. In recent decades, various studies have been carried out in the field of heat transfer within systems that have PCM by analyzing the latent energy stored in the system. These studies have included energy storage methods, latent heat storage materials, classification of PCMs, measurement methods, computational and heat transfer issues, thermal energy storage systems and their facilities, etc. The solar thermal energy storage system is one of the most widely used systems, and its use is expanding day by day.

This research focuses on enhancing the performance of solar thermal collectors and phase change materials (PCM) used for hot water production, with the aim of reducing our reliance on fossil fuels. While previous studies have separately optimized either the collectors or PCM, this study takes a comprehensive approach by combining both through a powerful and effective multiple-purpose evolutionary technique based on decomposition. This approach, widely recognized in engineering and resource planning, excels at handling multiple objectives simultaneously. The optimization algorithm targets two crucial parameters - energy discharge time (t_{PCM}) and net quantity of stored energy (Q_{net}) - which significantly impact the overall efficiency of the system. To further refine the optimization process, decision-making techniques and response surface methodology have been employed. This integrated approach holds tremendous potential for significantly improving the performance of solar thermal collectors and PCM storage tanks, leading to more efficient and cost-effective energy systems. The outcomes of this study not only contribute to advancing renewable energy technologies but also have the potential to guide future research and development efforts. By moving closer to a sustainable future, this research takes a meaningful step towards reducing our carbon footprint and creating a cleaner and greener world.

2. Methodology

2.1. Method

In a genuine system, there are a number of objectives that may need to be achieved simultaneously even when they are in conflict. A multiple-purpose evolutionary approach has been utilized in several researches to verify that the system has reached one of the many optimum stages. This approach has been applied in engineering problem-solving, work scheduling, regulation of resource storage, and various other areas, proving to be a straightforward and highly effective method [44]. A problem involving multiple optimization objectives is broken down into a number of single-objective optimizing sub-problems as part of the suggested method for solving it. These sub-problems are simultaneously solved using a population-based optimization method. A weight vector is assigned to each sub-problem, which is used to aggregate the objective functions into a single objective function. To ensure the stability of the solutions, we assume that certain conditions hold, including one-dimensional flow and a sky that behaves like a black substance for long irradiation of heating properties that are independent of temperature.

Thermal energy phase change materials can be stored in two forms sensible and latent heat energy. In sensible energy storage, thermal energy is stored in a solid or liquid object by increasing its temperature. The amount of sensible energy stored in the body depends on the temperature, specific heat capacity, and mass of the body. Thermal energy is stored latently by the body when the body changes its phase from solid to liquid liquid to gas or solid to solid. Phase change materials store energy in the form of latent heat of fusion. Heat storage is done in 3 ways phase change, In the first case, which is the phase change from solid to solid, it is not suitable because the heat transfer is very slow and small. In the second case, it is not practical to change the liquid to gas phase due to the need for high heat and the creation of a high-pressure volume of gas. However the phase change from solid to liquid is more suitable, and this feature exists in PCM phase change materials, which change from solid to liquid phase by absorbing heat at a constant temperature. These materials release energy almost at the same temperature as they absorb.

The objectives of the research are summarized as follows:

1- Evaluation of the performance of systematic effective committees.

2- Multi-objective optimization with the optimal design approach.

3- Increasing efficiency and reducing operating costs.

2.2. System description

Fig. 1 shows a schematic of the solar heat storage unit. This system consists of a flat plate solar collector, a water storage tank with disodium hydrogen phosphate as PCM inside the PCM tank, a pipeline, and several valves. The purified water is in pipes in the collector flow. When the sun shines, the temperature of the water inside the collector pipes increases. Then it goes into the water tank which is part of the PCM tank. Thermal energy is transferred from hot water to solid PCM, which has a melting temperature of 29 degrees Celsius. The heat transferred to the PCM changes it from the solid phase to the liquid phase. In this state, the water is ready for use. In the absence of solar energy, i.e. when the sun sets, the network water returns to the storage tank from the bypass pipeline and receives the energy stored during the day in the PCM liquid. In this process, the water temperature increases with the PCM phase change. After that, the water enters the pipelines for consumption and this thermodynamic cycle continues.

2.3. Governing equations

An energy analysis is conducted to assess the amount of thermal energy stored by the solar collector utilizing Phase Change Material (PCM). The beneficial energy that raises the temperature of the water in the collector is expressed by Q_u, which can be written at A_c level is as follows [45]:

$$\mathrm{Q_u = A_c F_R [S - U(T_c - T_a)]} \tag{1}$$

$$\mathrm{F_R} = \frac{\dot{\mathrm{m}}\mathrm{C_p}}{\mathrm{A_c U}}\left[1 - \mathrm{e}^{-\left[\frac{\mathrm{A_c U F'}}{\dot{\mathrm{m}}\mathrm{C_p}}\right]}\right] \tag{2}$$

This energy is equal to the difference between the energy received from the sun S(w/m2) and dissipated heating energy. F' is the collector efficiency factor which is calculated as below [45]:

$$\mathrm{F'} = \frac{\mathrm{U_t^{-1}}}{\mathrm{W[U_L(D + (W - D)F)] + \frac{1}{\pi D_i h_f}}} \tag{3}$$

where, U_t, which may be computed as follows, is the overall heat transfer coefficient [45]:

$$\mathrm{U_t} = \frac{\sigma(\mathrm{T_a + T_p})(\mathrm{T_a}^2 + \mathrm{T_p}^2)}{[\epsilon_p + 0/0425\mathrm{N}(1 - \epsilon_p)]^{-1} + \left[\frac{2\mathrm{N+f-1}}{\epsilon_g}\right] - \mathrm{N}} \tag{4}$$

In which N is the number of covering glasses, ε_p is the plane emission intensity, ε_g indicates the glass emission intensity (0/88), T_a is the ambience temperature, β is the collector gradient and h_w shows the coefficient of wind heat transfer.

In this regard 3, U1 is the overall loss coefficient of the collector, which can be obtained from reference [46].

Finally, the value of QL is obtained from Eq. (9):

$$Q_L = U_l(T_{in} - T_\circ) \tag{5}$$

Also, f is expressed as Eq. (6) [46].

$$\mathrm{f} = (1 + 0.089\mathrm{h_w} - 0.1166\mathrm{h_w}\epsilon_p) + (1 + 0.07866\mathrm{N}) \tag{6}$$

In this study, with the goal to enhance the effectiveness of the solar heating system, the following functions are considered for optimization [46]: the discharge time of PCM (t_{PCM}) and net stored energy within the PCM (Q_{net}).

$$\mathrm{Q_{net}} = \mathrm{Q}_u + \mathrm{Q_L} \tag{7}$$

The length of time that water may be maintained warm without solar energy is known as the discharge time of PCM:

$$\mathrm{t_{PCM}} = \frac{\mathrm{Q_{net}}}{\mathrm{Q}_u} \tag{8}$$

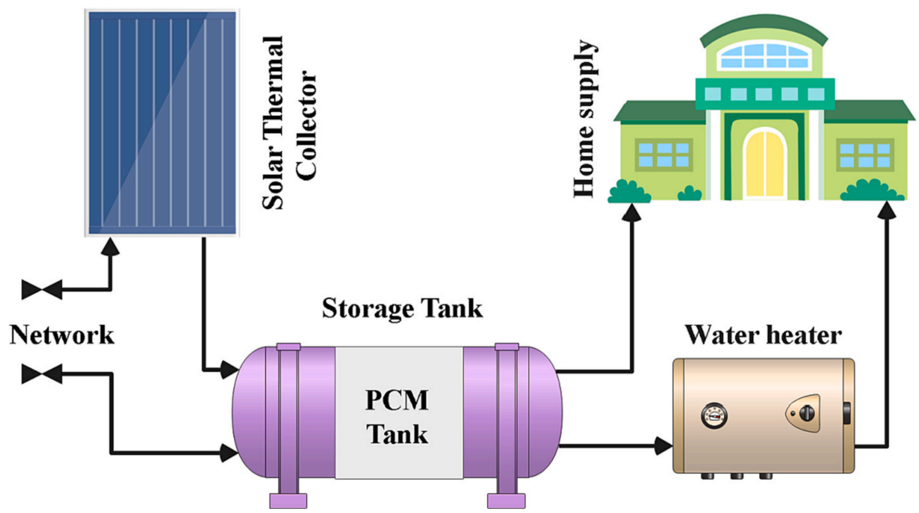

Fig. 1. Schematic of the proposed system.

2.4. Optimization

The priority of information is important in this type of optimization, and one of the methods of determining the priority of information is to determine the reference point. The reference point will determine the priority of the values to be entered into the target functions. The decomposition algorithm method can be used to determine the priority of information in multi-objective optimization of the evolutionary algorithm and it is shown as MOEA/D. MOEA/D is one of the easy and efficient methods of MOEA, which has been used in solving engineering problems, work planning, resource stock control, etc. This method decomposes a multi-objective optimization problem into several single-objective optimization problems. Then, from a population-based method, it solves these problems simultaneously. This algorithm is used in the present project by integrating its engineering problem.

The design parameters of the system are the inner diameter of the tubes (D), the area of collector (A_c), and Total energy stored in PCM minus the energy stored in water. In this study decision variables are the very design parameters of the system. The MOEA/D method divides the multiple-purpose problem into a number of simultaneously optimized scalar optimizing sub-problems. This method uses weight vectors that are defined for each sub-problem to combine the goal functions into a single objective function. An illustration of an optimization with multiple objectives problem is the following:

$$Min\ \mathrm{F(x)} = (\mathrm{f_1(x)}, \mathrm{f_2(x)}, \dots, \mathrm{f_m(x)})^{\mathrm{T}} \quad \text{Subject to } \mathrm{x} \in \mathrm{R^m} \tag{9}$$

In which R^m is the decision-making space with m number of objective functions with real values.

Table 1 introduces the decision variables and the range of constraints.

2.5. Response surface optimization method

In traditional experimental design methods, only one factor was considered as a variable, and other factors were placed at a fixed level, and in this method, the mutual effects between variables were not studied, and the full effects of the factors in the process could not be shown. Also, to conduct the research, a large number of experiments were needed, which led to an increase in time and cost, as well as an increase in the consumption of reagents and materials. To overcome this problem, the response surface method, or RSM was proposed for optimization studies. Response surface methodology is a set of statistical techniques and applied mathematics for building empirical models [47,48]. Fig. 2 shows the flowchart of the RSM method.

Table 1
Decision variables and the range of restrictions.

Decision variable	Lower bound	Upper bound
Diameter (m)	0.005	0.02
Area (m^2)	0.2	1
Ac (m^2)	0.26	0.38
Total energy stored in PCM minus the energy stored in water [kJ]	3100	3300

2.6. TOPSIS method

TOPSIS is a multi-indicator decision-making method for evaluating and prioritizing options based on criteria according to their distance from positive and negative ideals. The ideal solution is the one that increases the benefit criterion and decreases the cost criterion. The optimal option is the option that has the smallest distance from the ideal solution and at the same time the farthest distance from the negative ideal solution. In the ranking of options by the TOPSIS method, the options that are most similar to the ideal solution are ranked higher. In Fig. 3, the flowchart of the TOPSIS method is introduced [49,50].

2.7. LINMAP method

The LINMAP method is one of the multi-criteria MCDM decision-making methods used to determine the weights of criteria and sub-criteria and to rank options. In the LINMAP method, the number of m options with n indices is shown by m point vectors in an n-dimensional space, and it is assumed that the DM will choose the options with the smallest distance to the ideal point in this space [51,52].

2.8. AHP method

The Analysis Hierarchy Method (AHP) was developed by Saati in 1980. This technique is a powerful and flexible method in the category of multi-criteria decision-making methods that can solve complex problems at different levels. For this reason, it is called a hierarchy model because it is entered in the form of a tree model and levels [53,54].

2.9. Method

In this research, the storage system for solar heat is used. Energy

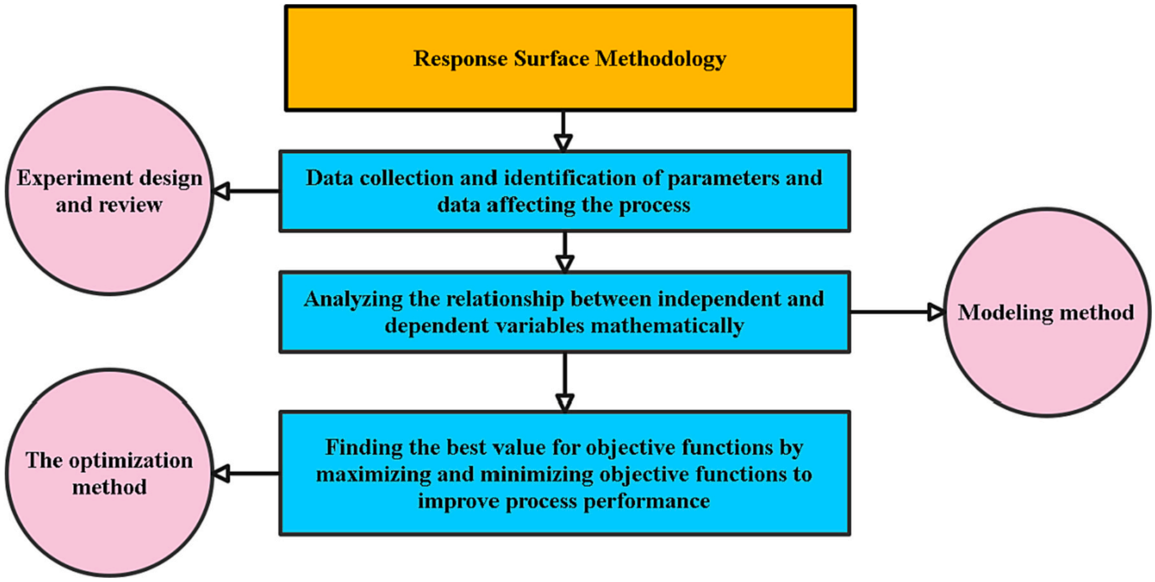

Fig. 2. Flowchart of RSM method.

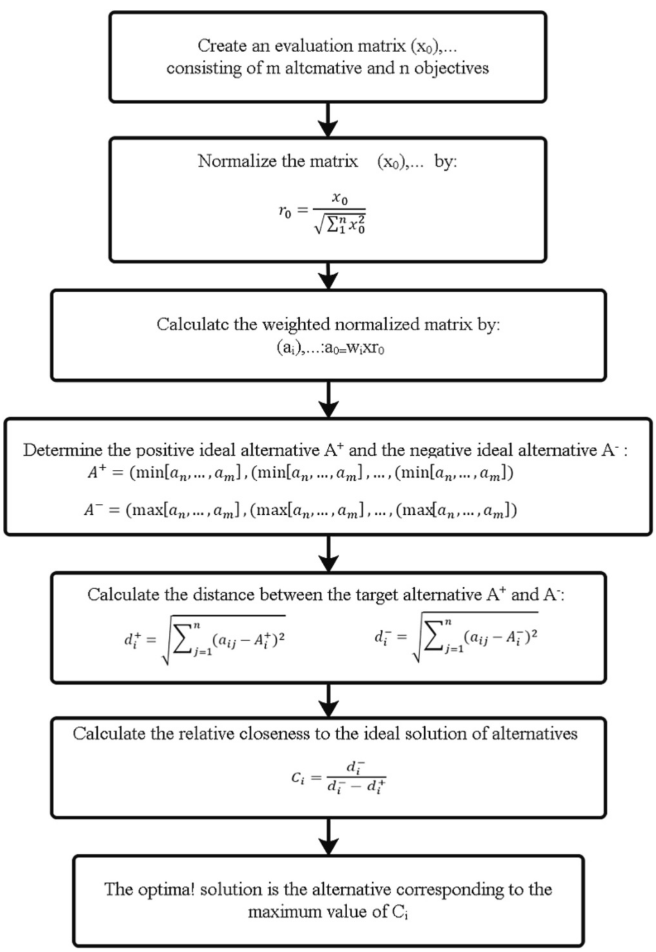

Fig. 3. Flowchart of the TOPSIS method.

analysis is used to evaluate the amount of thermal energy stored with the solar collector in PCM. Based on these studies and analyses, the system model is presented, system components are identified, and then programming is done accordingly. The efficiency of the mentioned system is evaluated depending on the committees, and the real goals of achieving the highest efficiency are subordinated. To achieve maximum efficiency, several goals must be met at the same time. These goals should be defined after clarifying the models. In general, it can be said that there are two main functions of energy (Qnet) and imaginary time of PCM (tpcm), the first one should be maximized and the second one should be minimized. For this purpose, the MOEA/D optimization algorithm is used. The research methodology is presented in the Fig. 4.

3. Results and discussion

3.1. Validation

Due to the novelty of the study, and to increase the accuracy of the modeling, the validation of the research has been done. In the research of Luo et al. in 2021 [55], the thermal modeling of the solar collector with the presence of PCM materials was investigated. The results of this comparison are presented in Table 2 and it shows a comparative analysis of the experimental value and the simulated value of the collector outlet temperature with the addition of the ambient temperature change curve, and as the results show, the modeling has good accuracy.

3.2. Results

Based on the ideal values of the stored energy within the PCM, Fig. 5 depicts the optimal values of the PCM's discharge time. As it can be elicited from the figure, when the conditions for the maximum time of energy discharge in the PCM are met (the left side of the diagram), the amount of the stored energy is at its lowest values. On the other hand, it is evident that the optimized conditions for the maximum amount of energy storage in the PCM leads to the least time of available energy at night-time. The population of the computational data in the conditions of Fig. 5 is equal to 500. Only two design parameters of diameter (D) and contact area (A) can be manipulated by the optimization. Based on the desired optimum objective, these parameters must be determined and used in the development of the system. In the conditions of Fig. 6, when 7 h of heating is required during night-time ($t_{PCM} = 7$ h), the optimum values of diameter and area are yielded as black points in Fig. 5. If 6 h of heating during night-time is considered ($t_{PCM} = 6$ h), optimal parameters are summarized in the blue stars.

The fluctuations of energy discharge as a function of tube inner diameter are depicted in Fig. 7. Other parameters are thought to be

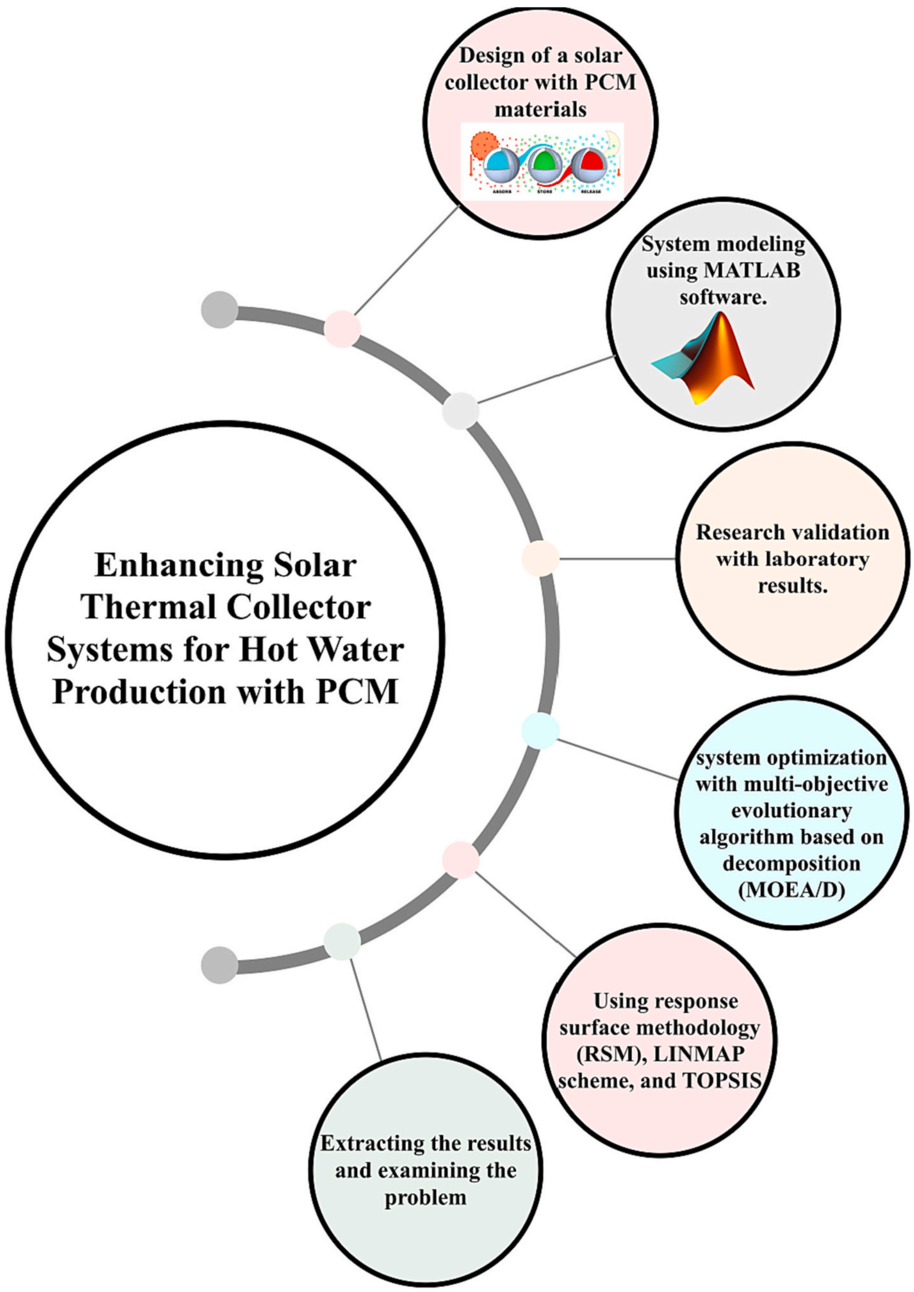

Fig. 4. Methodology.

Table 2
Validation.

Time	Temperature (°C)	
	Current study	Luo et al.
6:00	305	306
8:00	328.4	327.5
10:00	358.9	360
12:00	375	374
14:00	363	363
16:00	362	360
18:00	330.5	330

constants so that the impact of this parameter can be calculated. Table 3 provides a list of constant parameters.

Another decision variable is *Ac*, the effect of which on the variations in the energy discharge time and amount of stored energy in the PCM is shown in Fig. 8.

The blue line in Fig. 8 indicates the variations in t_{PCM} versus different values of storage tank area, while the red line exhibits the variations in Q_{net} versus storage tank area. As is observed, both the objective functions change linearly with storage tank contact area. The energy objective function of Q_{net} is more sensitive to the parameter of the area – compared to the objective function of t_{PCM}. Finally, the net energy stored in the PCM with respect to the tube diameter is estimated. Table 4 expresses the conditions in which other parameters are kept constant and the effect of diameter is investigated. The change span of the diameter varies from −0.02 to 0.005.

Fig. 9 represents the trend of variations in the net energy stored in the PCM (Q_{net}) for different values of the inner diameter. The trend is nonlinear and incremental. Therefore, by keeping the system conditions constant and increasing the diameter of the tube, the amount of energy stored in PCM also increases.

Q_{net} responds more sensitively to as the tube diameter increases. So,

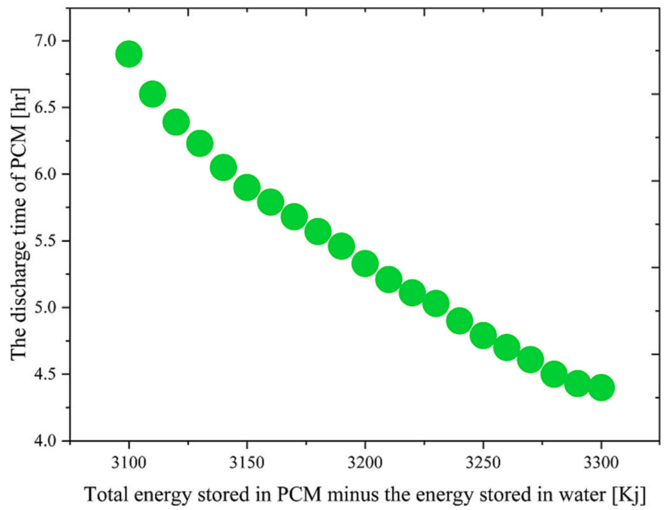

Fig. 5. MOEA/D optimal solution of the objective functions (net stored energy in PCM and energy discharge time).

in larger diameters, the amount of Q_{net} variations is more tangible. In addition to the aforementioned issues, the decision-making and RSM techniques have been utilized to optimize the responses. In Fig. 10, the optimal points that are obtained via the decision-making (DM) techniques all coincide at a single point. In case of 7 h of discharge time (t_{pcm} = 7 h), the energy stored in the PCM is equal to 3094 (kJ), whereas, for the discharge time of 6 h (t_{pcm} = 6 h) the amount of stored energy is equal to 3200 (kJ). It is worth mentioning the given values in Fig. 10 are obtained using the RSM method. Using the weighted Euclidean distance (di) on option Ai, the LINMAP technique assumes that the individual making the decision chooses the option from a collection of two potential options that is closest to the desired result. The existing scales are converted into unified ones using the weights-related variable (*Wj*), which makes it evident how important the indexes are. In LINMAP scheme, a number of *m* choices with *n* indexes through *m* vector points are shown in a three-dimensional space, and it is assumed that DM opts the choices with the least distance to the ideal point in this space. LINMAP is a point with the least distance from the ideal value. The TOPSIS technique, which Huang and Li first presented in 1981, remains one of the most effective multi-index methods for making decisions. This approach is predicated on the notion that the chosen option must be as far away as possible from both a positive ideal solution (the best state) and a negative ideal solution (the worst state). This model's fundamental logic identifies both the positive ideal solution and the negative ideal solution. The best option is one that raises the profit criteria and lowers the cost standard. The best alternative is the one that deviates most from both the desirable ideal scenario and the unpleasant desired outcome. In other words, when rating options using the TOPSIS technique, options that are closest to the optimal solution are given a higher

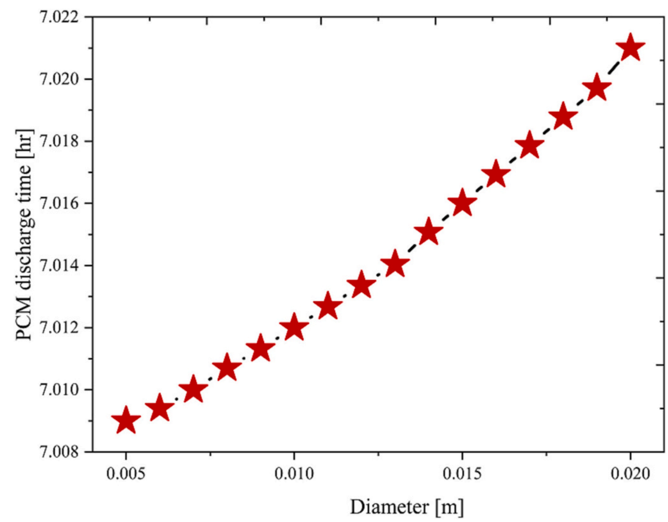

Fig. 7. Variations of energy discharge time versus the tube diameter.

Table 3
Conditions of variable diameter and fixed parameters.

Parameter	Unit	Value
Q_{net}	(kJ)	309732e3
$\dot{m}_w$	(kg/s)	0.009
A_c	(m^2)	0.287
m_{PCM}	(kg)	10
T_{in_water}	(K)	293.05

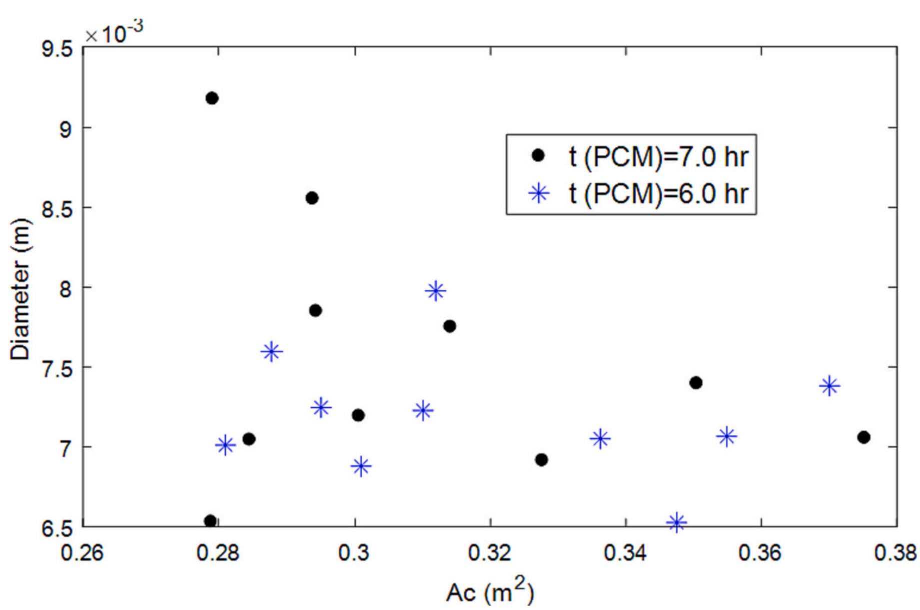

Fig. 6. Diameter and contact area parameters for the energy discharge times of 6 and 7 h.

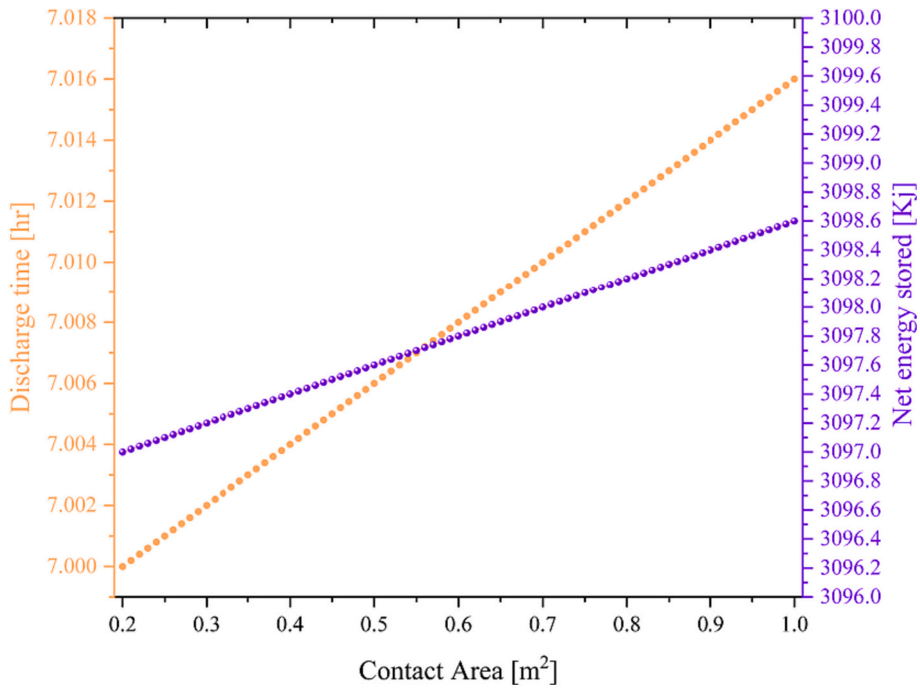

Fig. 8. Effects of the storage tank's contact area on the outcomes of objective functions.

Table 4
Fixed conditions for determining the effect of diameter on Q_{PCM}.

Parameter	Unit	Value
Q_{PCM}	(kJ)	309732e3
$\dot{m}_w$	(kg/s)	0.009
A_c	(m^2)	0.287
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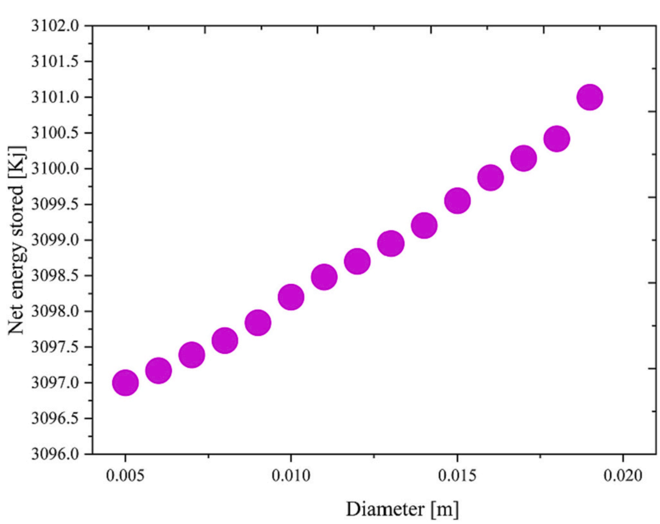

Fig. 9. Changes of the net energy stored in PCM versus the inner diameter of the tube.

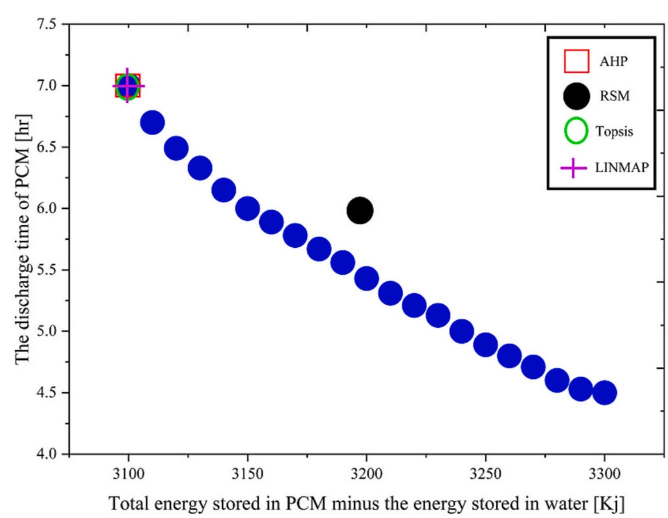

Fig. 10. Optimal findings of using decision-making and RSM techniques.

score. TOPSIS technique uses the worst value of the objective function in place of the ideal value. The ideal value is determined by taking the biggest value of the average created by the method known as AHP after standardizing the values of the goal functions.

Response surface methodology (RSM) is a collection of mathematical methods determining the relation between one or more response variables and several independent variables under study. This method was first developed by Box and Wilson in 1951. RSM has previously been utilized as a collection of practical statistical and mathematical tools for creating experimental models. RSM aims to optimize response (i.e. the output variable), which is influenced by several independent variables (i.e. input variables). In RSM scheme, the construction of the response surface models is a repetitive process. Once an approximate model is attained, the goodness of fit index is used to examine whether the response is satisfactory or not. If the response is not confirmed, the estimation of the process starts again and further experiments are then conducted.

It should be mentioned that in this research, the net amount of stored energy is shown by (Qnet) and it is investigated as the main objective function to increase the efficiency of the solar thermal collector. (tPCM) for the vital energy discharge time is considered as another objective function. The results show that the temperature value increases with the increase of time and considering that heat and temperature have a direct relationship, the amount of thermal energy increases with the increase of time and as a result, more energy will be stored.

In order to investigate the solid PCM materials, three classes of these materials (i.e., hybrid salts, paraffin, and fatty acids) were considered and one material from each of these classes was randomly chosen and studied. Effects of these materials with respect to the objective functions of Q_{net} and t_{PCM} are shown in Figs. 11 and 12. The characteristics of these materials are also shown in Table 5.

According to the results in Figs. 11 and 12, fatty acids and paraffin at a high melting temperature slightly increase the energy discharge time and net stored energy. Also, it is shown that at a high melting temperature, the hybrid salts, compared to paraffin and fatty acids, brings about a much greater increase in the energy discharge time and amount of net stored energy. Hybrid salts at a range of temperature from 0 to 150 degrees of centigrade have a high latent heat. Besides, they have wide applications due to their high heat storage capacity and density.

Fatty acids are mainly obtained from animal and vegetable sources, in addition, fatty acids are also produced from the oxidation of paraffin waxes.

Latent heat storage by phase change materials (PCM) can be used as a solution for the stability of solar equipment. The development of nanotechnology has led to the emergence of a group of high-performance nanostructured materials that have the ability to absorb thermal energy and release it when needed, and have gradually found industrial applications.

4. Conclusion

The thermal collector systems possess a substantial energy storage and supply capacity. This study focuses on utilizing phase change materials (PCMs) for storing solar energy during the daytime. PCMs absorb latent or sensible heat, enabling them to store solar energy effectively. The solar thermal collector's performance is optimized using the MOEA/D multi-objective optimization algorithm, considering the objective functions of energy discharge time (tPCM) and net stored energy in PCM (Qnet). The investigation reveals that increasing the inner diameter of the tube in the collector system leads to an increase in the energy discharge time of the PCM (tPCM). Consequently, the heat accessibility during the night is prolonged. The relationship between tPCM and tube diameter is nonlinear, with higher diameters exhibiting a more significant sensitivity to the energy discharge time compared to smaller diameters. Both objective functions, tPCM and Qnet, demonstrate linear changes concerning the contact area. The variations in these functions are directly related to the area parameter. The analysis of net stored energy in PCM (Qnet) for different tube diameters shows a nonlinear increasing trend. Keeping the system conditions constant while increasing the tube diameter results in higher energy storage in PCM. Qnet is more sensitive to alterations in the inner tube diameter, particularly in larger diameters. Consequently, precise selection of the tube diameter is crucial when employing large tanks in a collector system, as it significantly impacts the amount of stored energy (Qnet).

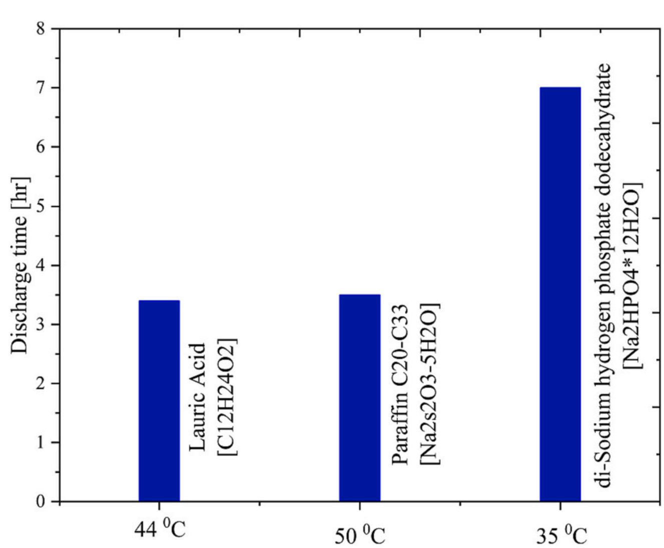

Fig. 11. Comparison of discharge time for different PCM materials.

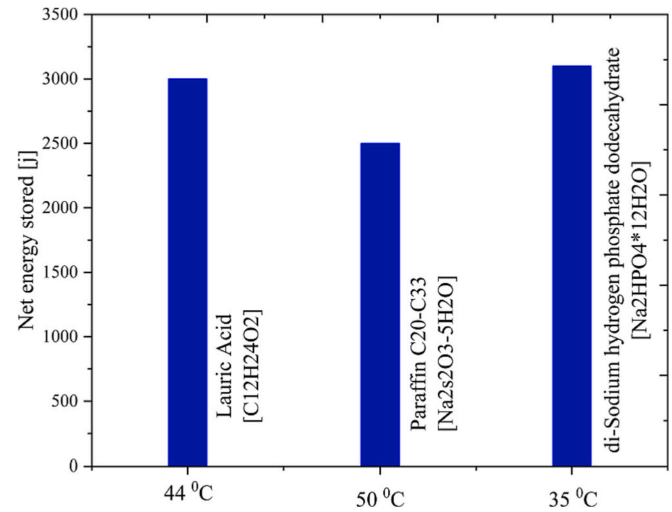

Fig. 12. Comparison of the stored energy for different PCM materials.

Table 5
Physical properties of different martials in this study [56,57].

	Parameter	Unit	Value
Hybrid salt (disodium hydrogen phosphate dodecahydrate)	Melting point	°C	35
	Density	kg/m^3	1522
	Solid-specific heat	kJ/kg K	1.55
	Liquid's specific heat	kJ/kg K	2.51
	Specific melting latent heat	kJ/kg	278.84
Paraffin (C20-C33 paraffin)	Melting point	°C	50
	Density	kg/m^3	912
	Liquid's specific heat	kJ/kg K	2.4
	Specific melting latent heat	kJ/kg	189
	Solid-specific heat	kJ/kg K	2.4
Fatty acid (uric acid)	Melting point	°C	44
	Density	kg/m^3	862
	Specific melting latent heat	kJ/kg	178
	Solid-specific heat	kJ/kg K	1.7
	Liquid's specific heat	kJ/kg K	2.3

CRediT authorship contribution statement

Ehsanolah Assareh: Conceptualization, Methodology, Software, Visualization, Investigation, Supervision, Writing- Reviewing and Editing, Data curation.
Amjad Riaz: Writing- Reviewing and Editing.
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Siamak Hoseinzadeh: Writing- Reviewing and Editing.
Mohammad Zaheri Abdehvan: Software, Visualization,

Investigation.

Sajjad Keykhah: Data curation, Software.

Tohid Jafarinejad: Data curation.

Rahim Moltames: Data curation.

Moonyong Lee: Supervision, Writing- Reviewing and Editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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